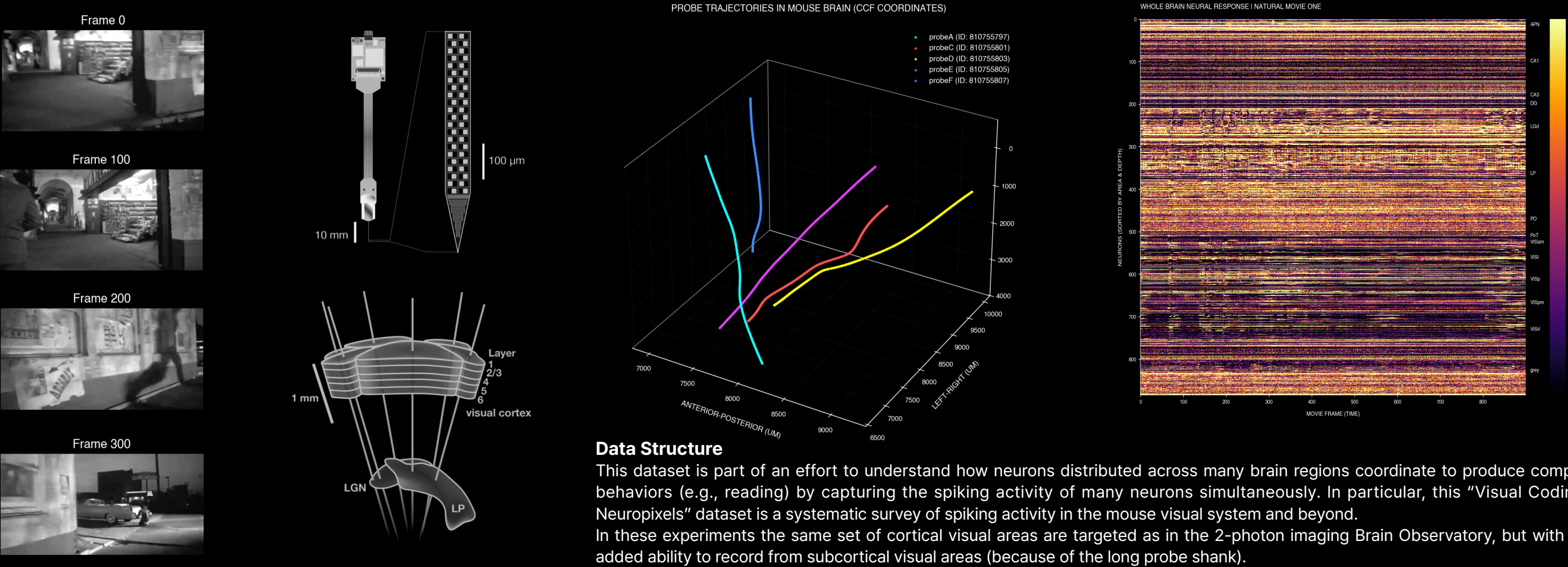


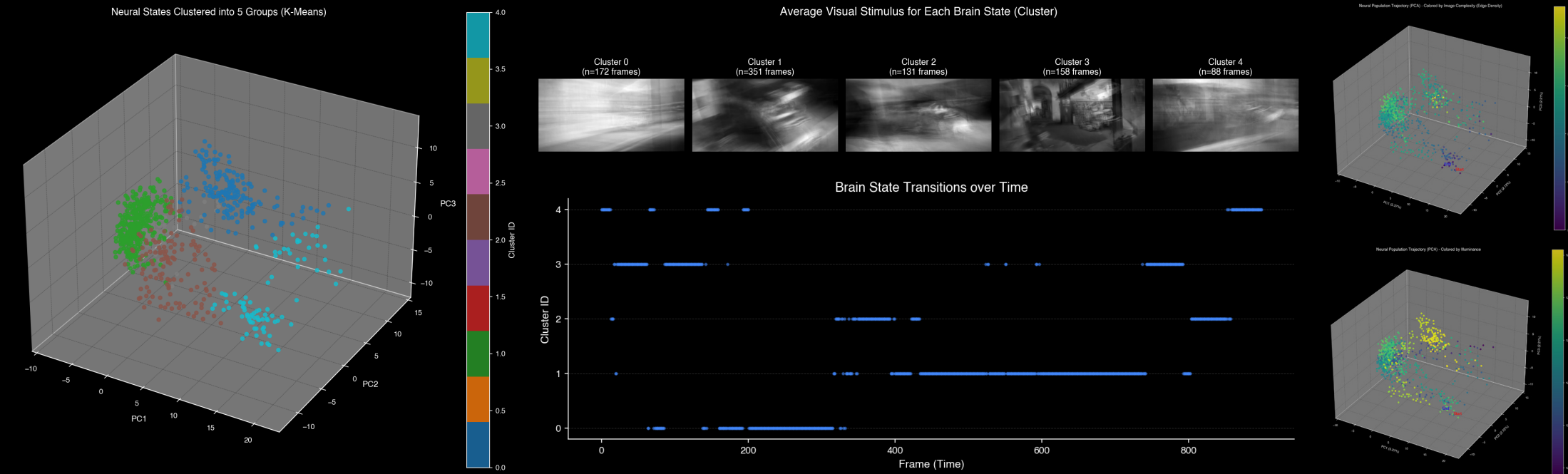
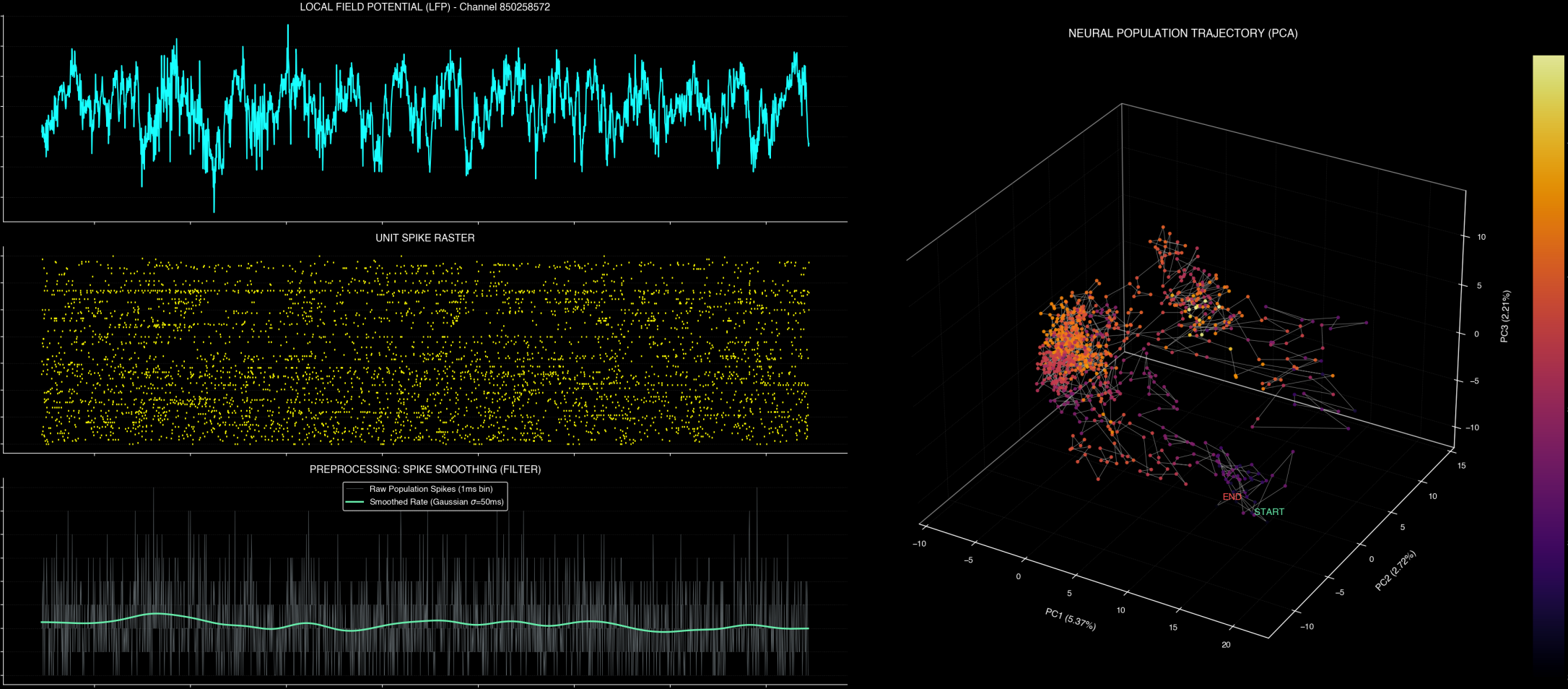
The Shape of Seeing the City

This project reveals how urban scenes unfold within high-dimensional neural geometry and aims to make these hidden structures accessible to the general public. Using spike data recorded while a mouse watches city videos, we transform population activity with PCA, UMAP, and persistent homology to uncover region-specific manifolds that reorganize visual complexity into distinct shapes. By turning these manifolds into a collectively explorable space, the project invites non-experts to map, interpret, and engage with neural structures that are typically reserved for scientific analysis.



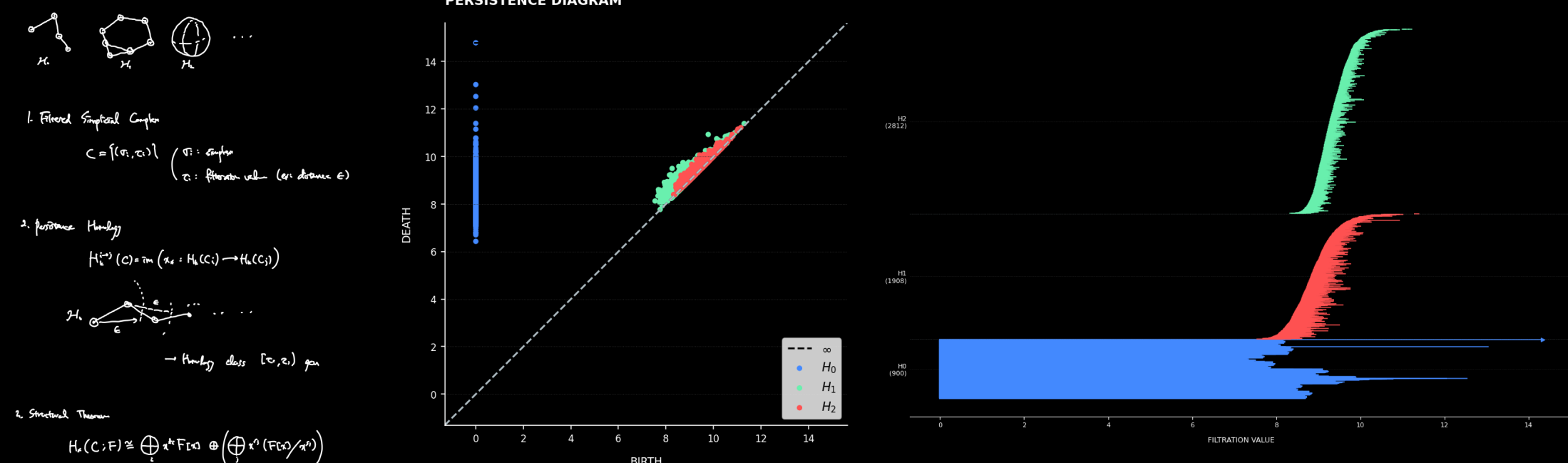
Continuous Representation of Spike

To obtain a continuous representation of neural population activity, each neuron's discrete spike train is first converted into a smooth firing-rate signal by convolving the binary spike sequence with a temporal filter, such as a Gaussian or exponential kernel. This filtering step removes the sparsity of the raw spike trains and captures the gradual changes in neural activation over time, producing a continuous-valued time series for every neuron. These filtered signals are then assembled into a population activity matrix, where each row corresponds to a neuron and each column corresponds to a time point.



K-mean Clustering

After applying PCA to the filtered population activity, we performed k-means clustering on the low-dimensional trajectories to group time points with similar neural population states. By averaging the neural responses within each cluster, we obtained a representative pattern for each stimulus segment, effectively summarizing the stimulus-driven dynamics while reducing noise and trial-to-trial variability. We also analyzed the visual stimulus itself, extracting features such as brightness and spatial complexity, and mapped these measurements onto the neural data to compare stimulus structure



UMAP and Neural Manifold

UMAP (Uniform Manifold Approximation and Projection) is a nonlinear dimensionality-reduction method that maps high-dimensional population activity into a low-dimensional space while preserving its local and global geometric structure. When applied to neural data, UMAP reveals the intrinsic manifold formed by population firing patterns. Because each brain region encodes information through distinct circuit dynamics and tuning properties, the resulting embeddings exhibit characteristic shapes: visual areas, which are strongly driven by the stimulus and show smooth, correlated responses, typically form clear, low-dimensional trajectories or branching structures; in contrast, regions such as CA1 often display more diffuse or higher-dimensional patterns that reflect internal state dynamics and mixed selectivity. These region-specific UMAP geometries provide insight into how different neural populations organize and transform sensory information.

